

E-Commerce Retention & Revenue Growth Analysis

Project Proposal · Business Data Mining & Decision Models (BDMDM)

GROUP MEMBERS

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Business Problem, Importance & Dataset

Protect existing customer value, cut revenue leakage, reduce dependence on two categories

39.9%

of orders cancelled / returned - \$4.73M gross unrealized

62.7%

of revenue from just 2 of 10 categories (concentration risk)

13.7%

of registered users never order; 31.7% of buyers one-time

THE OPPORTUNITY

- Revenue is flat (~\$529K/mo, 23-month window) - growth must come from protecting & reallocating existing value, not new demand.
- Reactivating buyers and reducing returns act on money already in the funnel: cheaper and faster than acquisition.
- Interventions must be targeted by customer value and risk - blanket discounts erode margin.
- Events, transactions, product and review data are complementary but must not be merged without validating their links.

DATASET - public e-commerce data, 6 linked tables (Jan 2024 - Nov 2025)

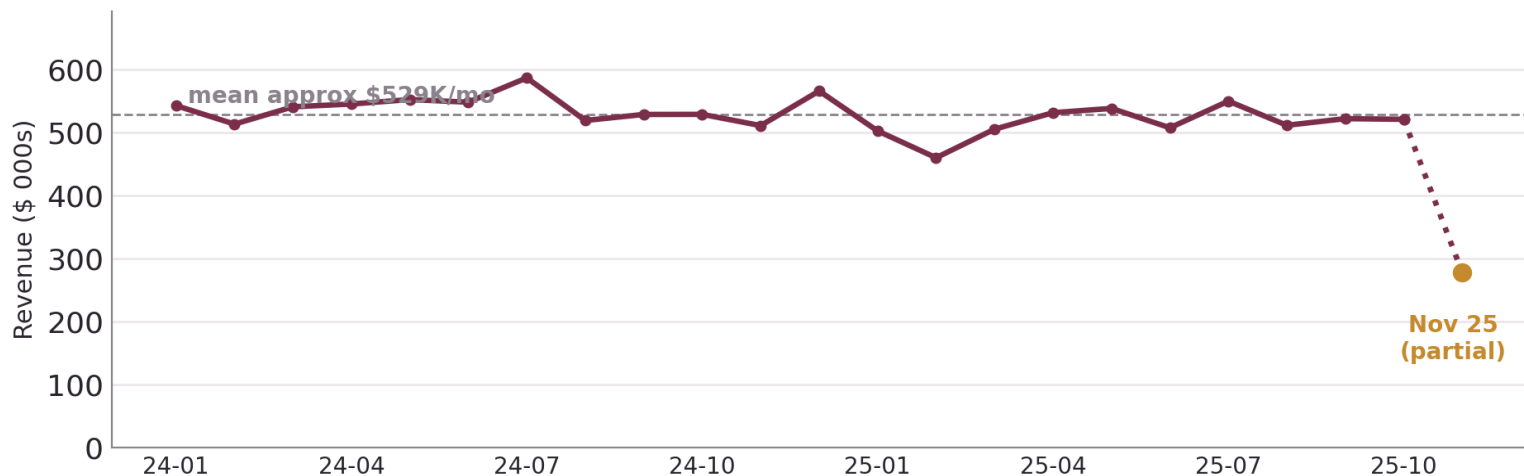
File	Rows	Role
users	10,000	customer master
products	2,000	catalog: category, brand, price
orders	20,000	header: status, date, amount
order_items	43,525	line items: qty, price, total
events	80,000	view / cart / wishlist / purchase
reviews	15,000	ratings + free text

DECISION PATHWAY

customer segment › inactivity / return risk › customer value › targeted offer › forecast budget › anomaly control

2 Descriptive Baseline - Customers, Orders & Revenue

23-month history, flat monthly revenue: the lever is reallocating value, not riding growth



\$11.92M

gross order value

\$595.93

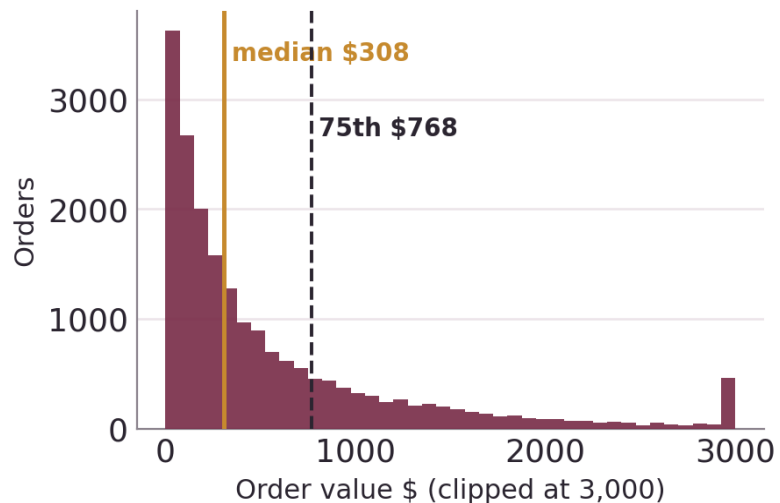
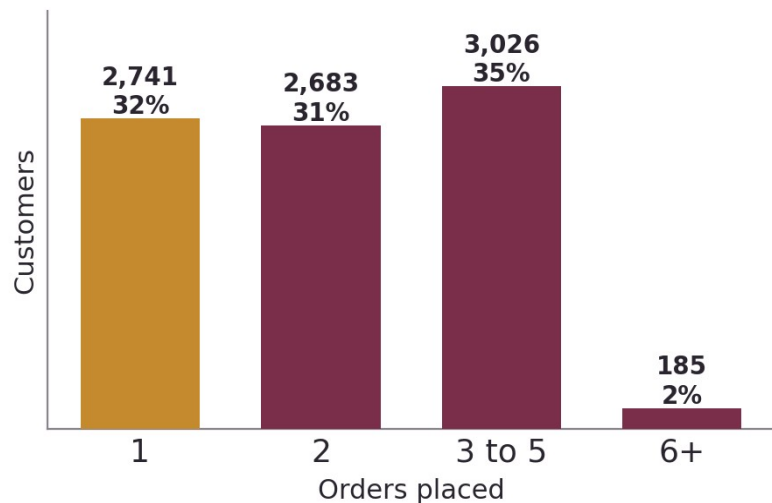
avg order value (AOV)

2.32

orders per buyer (median 2)

8,635

of 10,000 users purchased

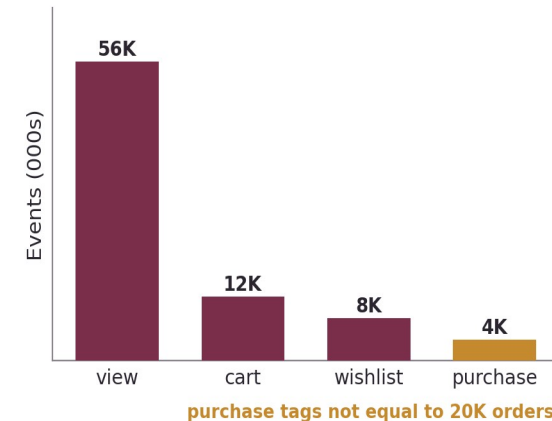
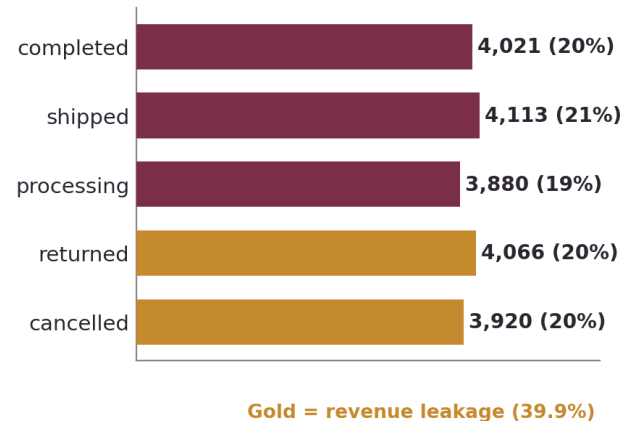
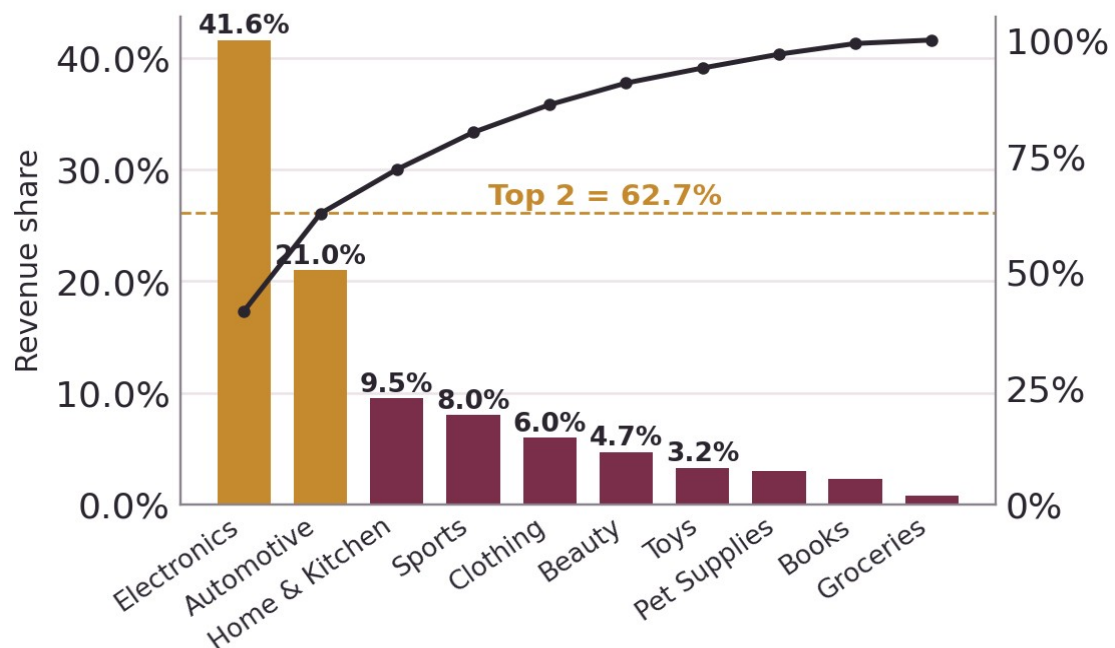


Read-outs

- Base skews to low-frequency buyers: 31.7% order once, only ~2% order 6+ times.
- Order value is long-tailed (median \$308, 75th \$768, max \$7,951).
- → Prioritise by value tier; the tail seeds anomaly detection.

3 Descriptive Baseline - Concentration & Engagement

Revenue concentrates in two categories; engagement events are not a verified purchase funnel



Two rating fields - keep separate

- products.rating: catalog score, float 2.5-4.9 (mean 3.68) → product similarity.
- reviews.rating: customer score, integer 1-5 (mean 3.54, n=15,000) → feedback analysis.
- Different constructs - never average or pool them.

Evidence boundary

- 4,006 event purchase-tags do NOT reconcile to 20,000 orders.
- No clean view → cart → purchase funnel; don't use event 'purchase' as transaction truth.
- Report engagement and transactions as separate analytical streams.

ALREADY VERIFIED

No basic hygiene needed

- 0 nulls, 0 duplicate rows
- All foreign keys resolve
- $\text{qty} \times \text{item_price} = \text{item_total}$ (all 43,525 rows)
- Order totals reconcile to line sums

SYNTHETIC-DATA CAVEAT

- Faker-style: @example.* emails, generated names, near-uniform category/gender/status.
- → Behavioural signal & real-world generalisability are limited; expectations tempered.

TARGETED PLAN

1 Fix the tenure anchor.

65% of first orders predate signup_date (median tenure - 107 days). Drop signup for tenure/cohorts; rebuild recency from first order_date.

2 Define the revenue rule.

Recognised = completed + shipped; exclude cancelled + returned (39.9%); report processing separately; run a status-sensitivity check.

3 Encode nominals as one-hot.

gender, order_status, event_type, category, brand - never label-encode into K-means / DBSCAN / kNN.

4 Keep IDs as strings; drop noise.

U/O/P/I ids stay identifiers (never numeric). Drop city (7,755 near-unique = an ID) and name / email (PII).

5 Engineer features.

Customer RFM from order_date (recency, frequency, recognised spend, AOV, category/brand breadth) + engagement counts; log1p on skewed money fields.

6 Separate event & transaction data.

Events as engagement predictors only - never as order ground truth.

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Proposed Models & Expected Outcomes

Interpretable, syllabus-aligned models to prioritise decisions - not to promise guaranteed uplift

Business question	Model(s)	Target / features	Realistic expected outcome
Who are our segments & who's valuable?	K-means (+ hierarchical / DBSCAN validation)	RFM + breadth + engagement, standardised	3-5 actionable cohorts; DBSCAN confirms whether dense natural clusters exist
Who will repeat / go inactive?	Logistic → decision tree / ensemble; class-imbalance handling	Repeat-purchase in window; leakage-safe RFM + engagement	Ranked risk list + drivers; expect weak signal (AUC ≈ 0.5) - stated as a data limit
What is each customer worth?	Linear + Ridge / Lasso	Future-period spend; historical spend, freq, AOV, recency	Relative value ranking; Lasso prunes drivers, Ridge stabilises correlated ones
What sells together / cold-start?	Association rules (Apriori) + content similarity	Category/brand baskets; category, brand, price, catalog rating	Cross-sell / bundle candidates by lift (≠ causal); similar-product list
Which transactions need review?	Robust z-score / IQR + Isolation Forest	total_amount, quantity, unit price, category norms	Transaction-risk queue; threshold from value exposure, not default 15%

Evaluation:

time-based split • same holdout across models • ROC-AUC / F1 (not raw accuracy) • holdout error for regression & forecast • clusters validated statistically + commercially • anomaly flags = investigation signals, not fraud. Not proposing RNN forecasting (23 monthly points, flat ~\$529K) - use classical smoothing / ARIMA if shown.